

A Technical Introduction to Natural language processing

Section 1

Apply the theory of conceptual metaphor, explained by Lakoff as "the understanding of one idea, in terms of another" which provides an idea of the intent of the author. For example, consider the English word big. When used in a comparison ("That is a big tree"), the author's intent is to imply that the tree is physically large relative to other trees or the authors experience. When used metaphorically ("Tomorrow is a big day"), the author's intent to imply importance. The intent behind other usages, like in "She is a big person", will remain somewhat ambiguous to a person and a cognitive NLP algorithm alike without additional information. Assign relative measures of meaning to a word, phrase, sentence or piece of text based on the information presented before and after the piece of text being analyzed, e.g., by means of a probabilistic context-free grammar (PCFG). The mathematical equation for such algorithms is presented in US Patent 9269353:

Section 2

Statistical NLP (1990s–present) Up until the 1980s, most natural language processing systems were based on complex sets of hand-written rules. Starting in the late 1980s, however, there was a revolution in natural language processing with the introduction of machine learning algorithms for language processing. This shift was influenced by increasing computational power (see Moore's law) and a decline in the dominance of Chomskyan linguistic theories (e.g. transformational grammar), whose theoretical underpinnings discouraged the sort of corpus linguistics that underlies the machine-learning approach to language processing.

Section 3

Where RMM is the relative measure of meaning token is any block of text, sentence, phrase or word N is the number of tokens being analyzed PMM is the probable measure of meaning based on a corpora d is the non zero location of the token along the sequence of N tokens PF is the probability function specific to a language Ties with cognitive linguistics are part of the historical heritage of NLP, but they have been less frequently addressed since the statistical turn during the 1990s. Nevertheless, approaches to develop cognitive models towards technically operationalizable frameworks have been pursued in the context of various frameworks, e.g., of cognitive grammar, functional grammar, construction grammar, computational psycholinguistics and cognitive neuroscience (e.g., ACT-R), however, with limited uptake in mainstream NLP (as measured by presence on major conferences of the ACL). More recently, ideas of cognitive

NLP have been revived as an approach to achieve explainability, e.g., under the notion of "cognitive AI". Likewise, ideas of cognitive NLP are inherent to neural models multimodal NLP (although rarely made explicit) and developments in artificial intelligence, specifically tools and technologies using large language model approaches and new directions in artificial general intelligence based on the free energy principle by British neuroscientist and theoretician at University College London Karl J. Friston.

Section 4

Lexical semantics What is the computational meaning of individual words in context? Distributional semantics How can we learn semantic representations from data? Named entity recognition (NER) Given a stream of text, determine which items in the text map to proper names, such as people or places, and what the type of each such name is (e.g. person, location, organization). Although capitalization can aid in recognizing named entities in languages such as English, this information cannot aid in determining the type of named entity, and in any case, is often inaccurate or insufficient. For example, the first letter of a sentence is also capitalized, and named entities often span several words, only some of which are capitalized. Furthermore, many other languages in non-Western scripts (e.g. Chinese or Arabic) do not have any capitalization at all, and even languages with capitalization may not consistently use it to distinguish names. For example, German capitalizes all nouns, regardless of whether they are names, and French and Spanish do not capitalize names that serve as adjectives. This task is also referred to as token classification. Sentiment analysis (see also Multimodal sentiment analysis) Sentiment analysis involves identifying and classifying the emotional tone expressed in text. This technique involves analyzing text to determine whether the expressed sentiment is positive, negative, or neutral. Models for sentiment classification typically utilize inputs such as word n -grams, Term Frequency-Inverse Document Frequency (TF-IDF) features, hand-generated features, or employ deep learning models designed to recognize both long-term and short-term dependencies in text sequences. The applications of sentiment analysis are diverse, extending to tasks such as categorizing customer reviews on various online platforms. Terminology extraction The goal of terminology extraction is to automatically extract relevant terms from a given corpus. Word-sense disambiguation (WSD) Many words have more than one meaning; we have to select the meaning which makes the most sense in context. For this problem, we are typically given a list of words and associated word senses, e.g. from a dictionary or an online resource such as WordNet. Entity linking Many words—typically proper names—refer to named entities; here we have to select the entity (a famous individual, a location, a company, etc.) which is referred to in context.

Section 5

1950s: The Georgetown experiment in 1954 involved fully automatic translation of more than sixty Russian sentences into English. The authors claimed that within three or five years, machine translation would be a solved problem. However, real progress was much slower, and after the ALPAC report in 1966, which found that ten years of research had failed to fulfill the expectations, funding for machine translation was dramatically reduced. Little further research in machine translation was conducted in America (though some research continued elsewhere, such as Japan and Europe) until the late 1980s when the first statistical machine translation systems were developed.

1960s: Some notably successful natural language processing systems developed in the 1960s were SHRDLU, a natural language system working in restricted "blocks worlds" with restricted vocabularies, and ELIZA, a simulation of Rogerian psychotherapy, written by Joseph Weizenbaum between 1964 and 1966. Despite using minimal information about human thought or emotion, ELIZA was able to produce interactions that appeared human-like. When the "patient" exceeded the very small knowledge base, ELIZA might provide a generic response, for example, responding to "My head hurts" with "Why do you say your head hurts?". Ross Quillian's successful work on natural language was demonstrated with a vocabulary of only twenty words, because that was all that would fit in a computer memory at the time.

1970s: During the 1970s, many programmers began to write "conceptual ontologies", which structured real-world information into computer-understandable data. Examples are MARGIE (Schank, 1975), SAM (Cullingford, 1978), PAM (Wilensky, 1978), TaleSpin (Meehan, 1976), QUALM (Lehnert, 1977), Politics (Carbonell, 1979), and Plot Units (Lehnert 1981). During this time, the first chatterbots were written (e.g., PARRY).

1980s: The 1980s and early 1990s mark the heyday of symbolic methods in NLP. Focus areas of the time included research on rule-based parsing (e.g., the development of HPSG as a computational operationalization of generative grammar), morphology (e.g., two-level morphology), semantics (e.g., Lesk algorithm), reference (e.g., within Centering Theory) and other areas of natural language understanding (e.g., in the Rhetorical Structure Theory). Other lines of research were continued, e.g., the development of chatterbots with Racter and Jabberwacky. An important development (that eventually led to the statistical turn in the 1990s) was the rising importance of quantitative evaluation in this period.